

Task- 3 Prompt Engineering Challenge

Poor Prompt	Effective Prompt
Write a Python program to find the largest number in a list.	Write a clean and well-structured Python program to find the largest number in a list. Use meaningful variable names, add inline comments, include a function with a docstring, handle empty lists as an edge case and provide sample input/output along with test cases.
Write Python code to search for an element in a list.	Write an optimized Python program to search for an element in a sorted list using Binary Search. Include proper comments, explain the time and space complexity, compare it with Linear Search and provide test cases with sample input/output.
Explain recursion.	Act as a teacher for university student. Explain recursion in Python for 4th year students. Use simple language, a real-life analogy, a step-by-step explanation, a basic factorial example with comments, a recursion tree, common mistakes, and a comparison with iteration.
Write documentation for a Student Management System.	Generate documentation for a Student Management System project. Include a professional README, project overview, features, installation steps, prerequisites, folder structure, usage examples, configuration details, contribution guidelines, license information. Use structured and well defined format.
Write Python code to remove duplicate elements from a list.	Write an optimized and clean Python program to remove duplicate elements from a list while preserving the original order. Follow PEP 8 standards, use meaningful variable names, include comments and docstrings, explain the algorithm, analyze time and space complexity, handle edge cases, and provide sample input/output along with test cases.